26900124 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours)

REGULAR EXAMINATION October 2026

Fifth SEMESTER

Discipline Specific Elective (DSE) - MG5DSECND301 - FOOD MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Archaea are: [U] / [CO1]

   a). Prokaryotes b). Eukaryotes

   c). unicellular organisms d). multicellular organisms

2. Which characteristic of sugar products generally restricts the growth of many microorganisms? [A] / [CO2]

   a). Low water activity b). High protein content

   c). High moisture content d). Neutral pH

3. Which practice is most likely to introduce microorganisms into fresh vegetables? [A] / [CO2]

a). Use of contaminated irrigation water
b). Proper cold storage

c). Hygienic handling
d). Washing with potable water

4. A bruised fruit usually spoils faster because the damaged tissue: [A] / [CO2]

a). Provides sites for microbial entry and growth
b). Completely prevents microbial growth

c). Reduces the available nutrients
d). Eliminates moisture from the fruit

5. "Bone taint" spoilage is primarily caused by: [U] / [CO3]

a). Aerobic *Pseudomonas* spp.
b). Yeasts

c). Moulds
d). Proteolytic *Clostridium* spp.

6. Sandiness defect in ice cream is due to: [U] / [CO3]

a). Bacterial growth
b). Lactose crystallization

c). Fungal spoilage
d). Proteolysis

7. Which processing step in poultry slaughter is most associated with cross-contamination via shared water? [U] / [CO3]

a). Stunning
b). Scalding

c). Grading
d). Freezing

8. Which chemical contaminant is often found in fish? [A] / [CO3]

a). Mercury
b). Antibiotics

c). Mold spores
d). Yeast cells

9. Which method is most effective in preventing microbial contamination of milk? [U] / [CO3]

a). Refrigeration
b). Freezing

c). Pasteurization
d). Drying

10. Spoilage of poultry is accelerated by: [U] / [CO3]

a). Refrigeration
b). Freezing

c). Room temperature storage
d). Vacuum packing

[1x10 = 10]

Part B

Short Answer

Answer any **5** questions

Each question carries **2** marks

11. Name the three categories of food based on perishability with example. [U] / [CO1]

12. Explain the role of moisture in the spoilage of stored cereals. [U] / [CO2]

13. What are osmophilic yeasts? Why are they important in sugar-product spoilage? [U] / [CO2]

14. Mention two methods to prevent milk spoilage. [U] / [CO3]

15. Name two common bacterial contaminants of meat. [U] / [CO3]

16. What is candling, and why is it used? [A] / [CO3]

17. Explain why glucose oxidase is important in glucose biosensors. [A] / [CO4]

[2x5 = 10]

Part C

Short Essay

Answer any **6** questions

Each question carries **5** marks

18. Define cross-contamination and how it can be prevented. [U] / [CO1]

19. Describe the sources of contamination of sugar and sugar products and explain the factors that influence microbial survival in these products. [U] / [CO2]

20. Describe the major sources and routes of microbial contamination of fresh fruits from production to handling. [U] / [CO2]

21. Explain microbial contamination of milk during milking and handling. [U] / [CO3]

22. Describe the various sources of contamination in canned foods at pre-processing, in-process, and post-processing stages. [U] / [CO3]

23. Describe the signs of spoilage in fish. [U] / [CO3]

24. Explain flat sour spoilage and thermophilic anaerobic (TA) spoilage of canned foods. [U] / [CO3]

25. Explain the advantages of SCP over conventional protein sources. [A] / [CO4]

[5x6 = 30]

Part D

Essay

Answer any **2** questions

Each question carries **10** marks

26.	Explain the bacterial growth curve and highlight its significance.	[U] / [CO1]
27.	Discuss in detail the sources, types, and routes of microbial contamination of cereals and cereal products from harvesting to storage, and explain suitable measures for controlling contamination.	[U] / [CO2]
28.	Explain the microbial production of lactic acid, including organisms, substrates, process and industrial applications	[U] / [CO4]
29.	Discuss organisms, substrates, and the process of microbial fat production.	[U] / [CO4]

[10x2 = 20]